Michael (Misha, Mykhailo) Rashkovetskyi

CCAPP Postdoctoral Fellow (cosmology)
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November 9, 2026
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Employment

- **The Ohio State University** — Columbus, OH, USA
  CCAPP Postdoctoral Fellow — *2025 – 2028*
  - Center for Cosmology and AstroParticle Physics
- **Harvard College Observatory** — Cambridge, MA, USA
  Postdoctoral Fellow — *2025*

Education

- **Harvard University** — Cambridge, MA, USA
  Ph.D. in Astronomy and Astrophysics — *2020 – 2025*
  M.A. in Astronomy and Astrophysics, in passing — *2022*
  - Thesis: *Enhancing the analysis of the large-scale structure of the Universe for cutting-edge cosmological surveys with two-point correlation function and beyond* (May 2025)
  - Advisor: Prof. Daniel Eisenstein
  - Center for Astrophysics | Harvard & Smithsonian
- **Tel Aviv University** — Tel Aviv-Yafo, Israel
  B.Sc. in Physics, Summa Cum Laude (GPA: 98/100) — *2019 – 2020*
  - Raymond & Beverly Sackler School of Physics & Astronomy
  - Advisor: Dr. Omer Bromberg
- **Moscow Institute of Physics and Technology** — Dolgoprudny, Russia
  B.Sc. in Applied Mathematics and Physics, unfinished — *2015 – 2018*
  - Department of General and Applied Physics
  - Advisor: Prof. Vasily Beskin
- **Richelieu Lyceum** — Odesa, Ukraine
  High school, specialization in physics — *2010 – 2015*

First-author publications

4. M. **Rashkovetskyi**, D. J. Eisenstein, *et al.*, "Clustering of DESI galaxies split by thermal Sunyaev-Zeldovich effect", *The Open Journal of Astrophysics* **8** October (2025)a 46033, arXiv:2508.20904
3. M. **Rashkovetskyi**, D. Forero-Sánchez, A. de Mattia, D. J. Eisenstein, N. Padmanabhan, H. Seo, A. J. Ross, *et al.*, "Semi-analytical covariance matrices for two-point correlation function for DESI 2024 data", J. Cosmology Astropart. Phys. **2025** January (2025)b 145, arXiv:2404.03007

2. M. **Rashkovetskyi**, D. J. Eisenstein, *et al.*, "Validation of semi-analytical, semi-empirical covariance matrices for two-point correlation function for early DESI data", MNRAS **524** September (2023) 3894–3911, arXiv:2306.06320

1. M. **Rashkovetskyi**, J. B. Muñoz, D. J. Eisenstein, and C. Dvorkin, "Small-scale clumping at recombination and the Hubble tension", Phys. Rev. D **104** November (2021) 103517, arXiv:2108.02747

First-tier author publications

11. N. Sanders, H. Seo, M. **Rashkovetskyi**, U. Andrade, D. Valcin, E. Paillas, P. McDonald, *et al.*, "A Unified Tracer Analysis of DESI DR2 Baryon Acoustic Oscillations", August 2026, arXiv:2608.27830

10. E. A. Zaborowski, P. Taylor, K. Honscheid, A. Cuceu, A. de Mattia, A. Krolewski, M. **Rashkovetskyi**, A. J. Ross, C. To, *et al.*, "H_0 without the sound horizon (or supernovae): A 2% measurement in DESI DR1", J. Cosmology Astropart. Phys. **2026** April (2026) 004, arXiv:2510.19149

9. D. Valcin, M. **Rashkovetskyi**, H. Seo, F. Beutler, P. McDonald, A. de Mattia, A. J. Rosado-Marín, A. J. Ross, N. Padmanabhan, *et al.*, "Combined tracer analysis for DESI 2024 BAO", J. Cosmology Astropart. Phys. **2026** April (2026) 058, arXiv:2508.05467

8. U. Andrade, E. Paillas, J. Mena-Fernández, Q. Li, A. J. Ross, S. Nadathur, M. **Rashkovetskyi**, A. Pérez-Fernández, H. Seo, N. Sanders, O. Alves, X. Chen, N. Deiosso, A. de Mattia, M. White, *et al.*, "Validation of the DESI DR2 measurements of baryon acoustic oscillations from galaxies and quasars", Phys. Rev. D **112** October (2025) 083512, arXiv:2503.14742

7. D. Forero-Sánchez, M. **Rashkovetskyi**, O. Alves, *et al.*, "Analytical and EZmock covariance validation for the DESI 2024 results", J. Cosmology Astropart. Phys. **2025** April (2025) 055, arXiv:2411.12027

6. J. Yu, A. J. Ross, A. Rocher, O. Alves, A. de Mattia, D. Forero-Sánchez, J.-P. Kneib, A. Krolewski, T. Lan, M. **Rashkovetskyi**, *et al.*, "ELG spectroscopic systematics analysis of the DESI Data Release 1", J. Cosmology Astropart. Phys. **2025** January (2025) 126, arXiv:2405.16657

5. J. Mena-Fernández, C. Garcia-Quintero, S. Yuan, B. Hadzhiyska, O. Alves, M. **Rashkovetskyi**, *et al.*, "HOD-dependent systematics for luminous red galaxies in the DESI 2024 BAO analysis", J. Cosmology Astropart. Phys. **2025** January (2025) 133, arXiv:2404.03008

4. C. Garcia-Quintero, J. Mena-Fernández, A. Rocher, S. Yuan, B. Hadzhiyska, O. Alves, M. **Rashkovetskyi**, *et al.*, "HOD-dependent systematics in Emission Line Galaxies for the DESI 2024 BAO analysis", J. Cosmology Astropart. Phys. **2025** January (2025) 132, arXiv:2404.03009

3. J. Moon, D. Valcin, M. **Rashkovetskyi**, C. Saulder, *et al.*, "First detection of the BAO signal from early DESI data", MNRAS **525** November (2023) 5406–5422, arXiv:2304.08427

2. E. M. Novoselov, V. S. Beskin, A. K. Galishnikova, M. M. **Rashkovetskyi**, and A. V. Biryukov, "Orthogonal pulsars as a key test for pulsar evolution", MNRAS **494** April (2020) 3899–3911, arXiv:2004.03211

1. V. S. Beskin, A. K. Galishnikova, E. M. Novoselov, A. A. Philippov, and M. M. **Rashkovetskyi**, "So how do radio pulsars slow-down?", in "Journal of Physics Conference Series", vol. 932, p. 012012. December 2017

Collaboration publications

20. DESI Collaboration *et al.*, "DESI DR2 Results IV: Alcock-Paczyński Measurements from the Lyman Alpha Forest and Cosmological Constraints", July 2026, arXiv:2607.27410

19. C. Garcia-Quintero *et al.*, "Cosmological implications of DESI DR2 BAO measurements in light of the latest ACT DR6 CMB data", Phys. Rev. D **112** October (2025) 083529, arXiv:2504.18464

18. DESI Collaboration *et al.*, "DESI DR2 results. II. Measurements of baryon acoustic oscillations and cosmological constraints", Phys. Rev. D **112** October (2025)a 083515, arXiv:2503.14738

17. DESI Collaboration *et al.*, "DESI DR2 results. I. Baryon acoustic oscillations from the Lyman alpha forest", Phys. Rev. D **112** October (2025)b 083514, arXiv:2503.14739

16. K. Lodha *et al.*, "Extended dark energy analysis using DESI DR2 BAO measurements", Phys. Rev. D **112** October (2025) 083511, arXiv:2503.14743

15. W. Elbers *et al.*, "Constraints on neutrino physics from DESI DR2 BAO and DR1 full shape", Phys. Rev. D **112** October (2025) 083513, arXiv:2503.14744

14. DESI Collaboration *et al.*, "Data Release 1 of the Dark Energy Spectroscopic Instrument", AJ **171** May (2026) 285, arXiv:2503.14745

13. X. Chen, Z. Ding, E. Paillas, *et al.*, "Extensive analysis of reconstruction algorithms for DESI 2024 baryon acoustic oscillations", J. Cosmology Astropart. Phys. **2026** May (2026) 001, arXiv:2411.19738

12. DESI Collaboration *et al.*, "DESI 2024 II: sample definitions, characteristics, and two-point clustering statistics", J. Cosmology Astropart. Phys. **2025** July (2025)a 017, arXiv:2411.12020

11. DESI Collaboration *et al.*, "DESI 2024 V: Full-Shape galaxy clustering from galaxies and quasars", J. Cosmology Astropart. Phys. **2025** September (2025)b 008, arXiv:2411.12021

10. DESI Collaboration *et al.*, "DESI 2024 VII: cosmological constraints from the full-shape modeling of clustering measurements", J. Cosmology Astropart. Phys. **2025** July (2025)c 028, arXiv:2411.12022

9. A. Pérez-Fernández *et al.*, "Fiducial-cosmology-dependent systematics for the DESI 2024 BAO analysis", J. Cosmology Astropart. Phys. **2025** January (2025) 144, arXiv:2406.06085

8. U. Andrade *et al.*, "Validating the galaxy and quasar catalog-level blinding scheme for the DESI 2024 analysis", J. Cosmology Astropart. Phys. **2025** January (2025) 128, arXiv:2404.07282

7. DESI Collaboration *et al.*, "DESI 2024 III: baryon acoustic oscillations from galaxies and quasars", J. Cosmology Astropart. Phys. **2025** April (2025)a 012, arXiv:2404.03000

6. DESI Collaboration *et al.*, "DESI 2024 IV: Baryon Acoustic Oscillations from the Lyman alpha forest", J. Cosmology Astropart. Phys. **2025** January (2025)b 124, arXiv:2404.03001

5. DESI Collaboration *et al.*, "DESI 2024 VI: cosmological constraints from the measurements of baryon acoustic oscillations", J. Cosmology Astropart. Phys. **2025** February (2025)c 021, arXiv:2404.03002

4. E. Paillas, Z. Ding, X. Chen, *et al.*, "Optimal reconstruction of baryon acoustic oscillations for DESI 2024", J. Cosmology Astropart. Phys. **2025** January (2025) 142, arXiv:2404.03005

3. S. F. Chen, C. Howlett, *et al.*, "Baryon acoustic oscillation theory and modelling systematics for the DESI 2024 results", MNRAS **534** October (2024) 544–574, arXiv:2402.14070

2. DESI Collaboration *et al.*, "Validation of the Scientific Program for the Dark Energy Spectroscopic Instrument", AJ **167** February (2024)a 62, arXiv:2306.06307

1. DESI Collaboration *et al.*, "The Early Data Release of the Dark Energy Spectroscopic Instrument", AJ **168** August (2024)b 58, arXiv:2306.06308

Selected open source contributions (https://github.com/misharash)

- RascalC — C++ & Python
  Fast semi-analytic covariance matrices library/program — *2022 – 2026*
- RascalC scripts — Python
  DESI covariance matrix pipeline for 2-point correlation function (scripts) — *2024 – 2026*
- pycorr — Python
  Library for 2-point correlation function estimation — *2024 – 2025*

Seminar talks

- **Georgia Tech Center for Relativistic Astrophysics Seminar** — Atlanta, GA, USA
  Spectral survey instrument reveals notes of discord with cosmic sound waves: DESI overview, BAO measurements and their cosmological implications — *November 5, 2026*
- **CCAPP Fellows Symposium 2026** — Columbus, OH, USA
  Validation of semi-analytic covariance matrices for DESI DR2 clustering measurements — *September 30, 2026*
- **Ohio Cosmic Connections** — Cincinnati, OH, USA
  Clustering of DESI galaxies split by thermal Sunyaev-Zeldovich effect — *May 4, 2026*
- **University of Michigan Cosmology Group Seminar** — Ann Arbor, MI, USA
  Semi-analytical covariance matrices for DESI BAO measurements, and more information from tSZ-split clustering — *April 10, 2026*
- **Ohio University Astro Seminar** — Athens, OH, USA
  Enhancing the analysis of the large-scale structure of the Universe — *October 8, 2025*
- **CCAPP Fellows Symposium 2025** — Columbus, OH, USA
  Clustering of DESI galaxies split by thermal Sunyaev-Zeldovich effect — *September 17, 2025*
- **KIPAC Tea (Stanford / SLAC)** — Menlo Park, CA, USA
  Fast semi-analytical covariance matrices for 2-point correlation functions — *May 10, 2024*
- **ITC Luncheon** — Cambridge, MA, USA
  Semi-analytic covariance matrices for 2PCF of DESI galaxies (on YouTube) — *September 28, 2023*

Contributed conference talks

- **New Frontiers in Cosmology 2026** — A Coruña, Spain
  tSZ-split clustering of DESI galaxies — *August 13, 2026*
- **APS Global Physics Summit** — Denver, CO, USA
  Clustering of DESI galaxies split by thermal Sunyaev-Zeldovich effect — *March 17, 2026*

- **28th International Conference on Particle Physics & Cosmology** — Pittsburgh, PA, USA
  Clustering of DESI galaxies split by thermal Sunyaev-Zeldovich effect — *October 16, 2025*
- **APS Global Physics Summit** — Anaheim, CA, USA
  Clustering of DESI LRG selected based on ACT DR6 + Planck tSZ map — *March 18, 2025*
- **IX Essential Cosmology for the Next Generation** — Playa del Carmen, Mexico
  Clustering of DESI galaxies selected based on ACT thermal SZ map — *December 5, 2024*
- **Cosmology from Home 2023** — online
  First Detection of the BAO Signal from Early DESI Data (on YouTube) — *July 4, 2023*
  Co-presenter with Christoph Saulder, Jeongin Moon and David Valcin
- **APS April Meeting** — Minneapolis, MN, USA
  First Detection of the BAO Signal from Early DESI Data — *April 18, 2023*

Poster presentations

- **Arizona Winter School on Multi-Probe Cosmology** — Tucson, AZ, USA
  Clustering of DESI galaxies split by thermal Sunyaev-Zeldovich effect — *January 12 – 15, 2026*
  Second best poster prize
- **Frontiers in Cosmology and Gravitational Physics** — Portsmouth, UK
  Fast semi-analytical covariance matrices for 2PCF of galaxies and quasars — *May 20 – 23, 2024*
- **VIII Essential Cosmology for the Next Generation** — Playa del Carmen, Mexico
  `RascalC`*: Empirical 2PCF Covariance Matrices without Mocks* — *Nov 30 – Dec 3, 2022*
- **CMB-S4 Summer Meeting** — online
  Small-scale Clumping at Recombination and the Hubble Tension — *August 9 – 13, 2021*
- **CMB-S4 Spring Meeting** — online
  Hubble Tension with Small-Scale Clumping — *March 8 – 12, 2021*
- **Physics of Neutron Stars - 2017** — Saint-Petersburg, Russia
  On the light-curve anomalies of radio pulsars — *July 10 – 14, 2017*

Other meetings

- **DESI December** — Tucson, AZ, USA
  DESI Collaboration — *December 8 – 11, 2025*
- **DESI July** — Berkeley, CA, USA
  DESI Collaboration — *July 8 – 11, 2025*
- **DESI December** — Cancún, Mexico
  DESI Collaboration — *December 10 – 13, 2024*
- **DESI July** — Marseille, France
  DESI Collaboration — *July 9 – 12, 2024*
- **Fundamental Physics from Future Spectroscopic Surveys** — Berkeley, CA, USA
  Lawrence Berkeley National Laboratory — *May 6 – 8, 2024*
- **DESI December** — Waikoloa, HI, USA
  DESI Collaboration — *December 11 – 14, 2023*
- **DESI July** — Durham, UK
  DESI Collaboration — *July 17-21, 2023*

- **2nd Michigan Cosmology Summer School** — Ann Arbor, MI, USA
  University of Michigan — *June 5 – 9, 2023*
- **DESI December** — Cancún, Mexico
  DESI Collaboration — *December 5 – 9, 2022*
- **DESI June** — Berkeley, CA, USA
  DESI Collaboration — *June 21 – 24, 2022*
- **15th School of Modern Astrophysics** — Dolgoprudny, Russia
  Moscow Insitute of Physics and Technology — *July 1 – 12, 2019*
- **13th School of Modern Astrophysics** — Dolgoprudny, Russia
  Moscow Insitute of Physics and Technology — *July 3 – 21, 2017*
- **International School of Subnuclear Physics - 2017** — Erice, Italy
  "Ettore Majorana" Foundation and Centre for Scientific Culture — *June 14 – 23, 2017*
- **International school on particles, fields and strings** — Moscow, Russia
  National Research University "High School of Economics" — *April 17 – 24, 2017*
- **Astronomical practice** — Nizhniy Arkhyz, Russia
  Special Astrophysical Observatory — *June 25 – July 2, 2016*

Outreach and teaching

- **DESI redshift-space distortions animations** — with Claire Lamman+
  Early version used in the press-release and following news articles — *Nov 19, 2024 – Mar 2, 2025*
- **Astronomy 200: Radiative Processes in Astrophysics** — Harvard University
  Teaching Fellow — *Fall 2023*
- **Astronomy 201: Astrophysical Fluids & Plasmas** — Harvard University
  Teaching Fellow — *Spring 2023*
- **Astronomy 130: Introduction to Cosmology** — Harvard College
  Teaching Fellow — *Fall 2022*

Awards, grants and honors

Dean's Certificate in Recognition of Outstanding Academic Achievements (TAU)	2019 – 2020
Stipend for excellent students of MIPT in the name of A. Abramov	2016 – 2017
International Physics Olympiad, bronze medal	Mumbai, 2015
International Physics Olympiad, silver medal	Astana, 2014

Languages

- **Russian:** native
- **Ukrainian:** fluent
- **English:** advanced
- **Hebrew:** advanced
- **German:** intermediate